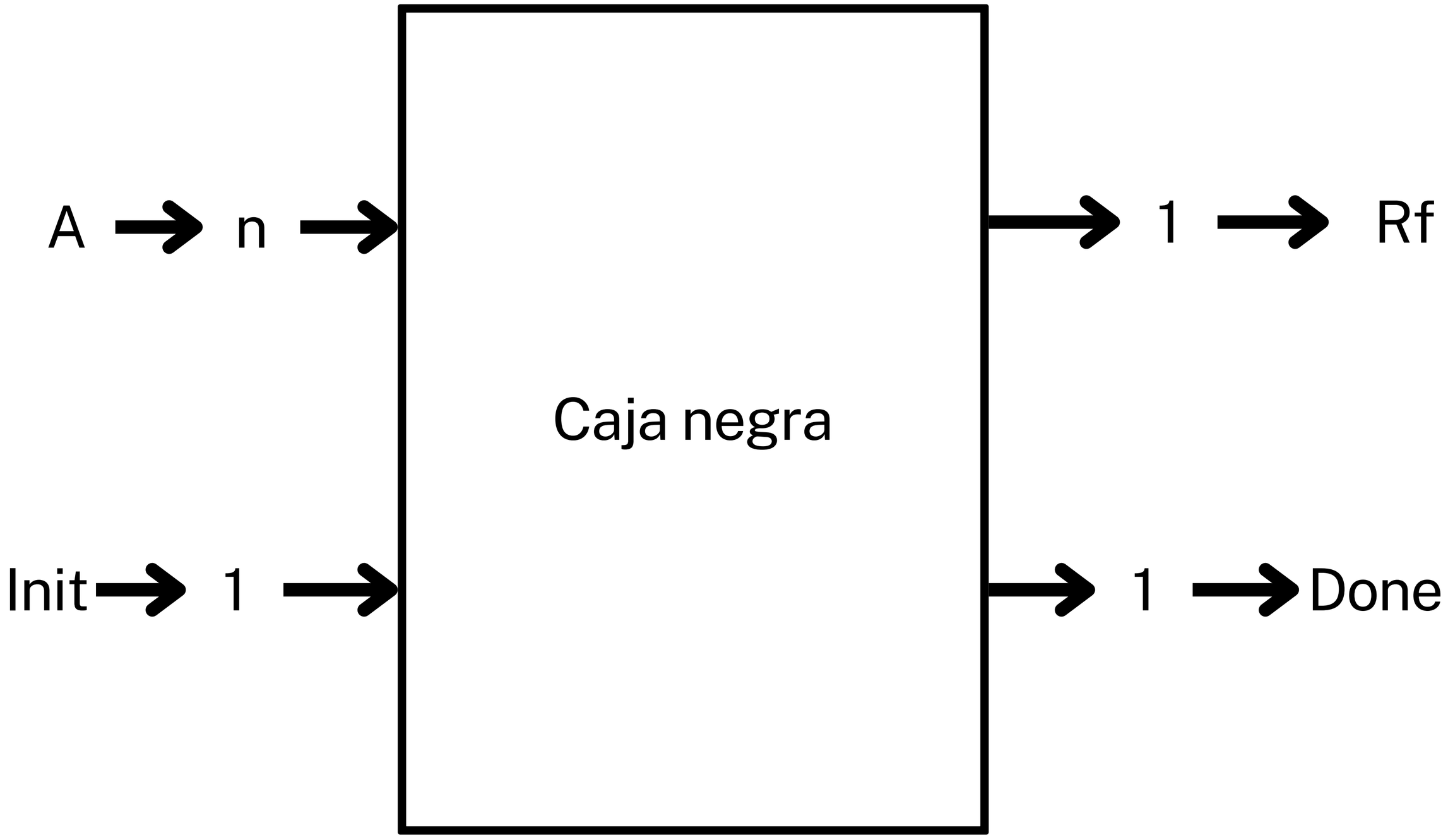
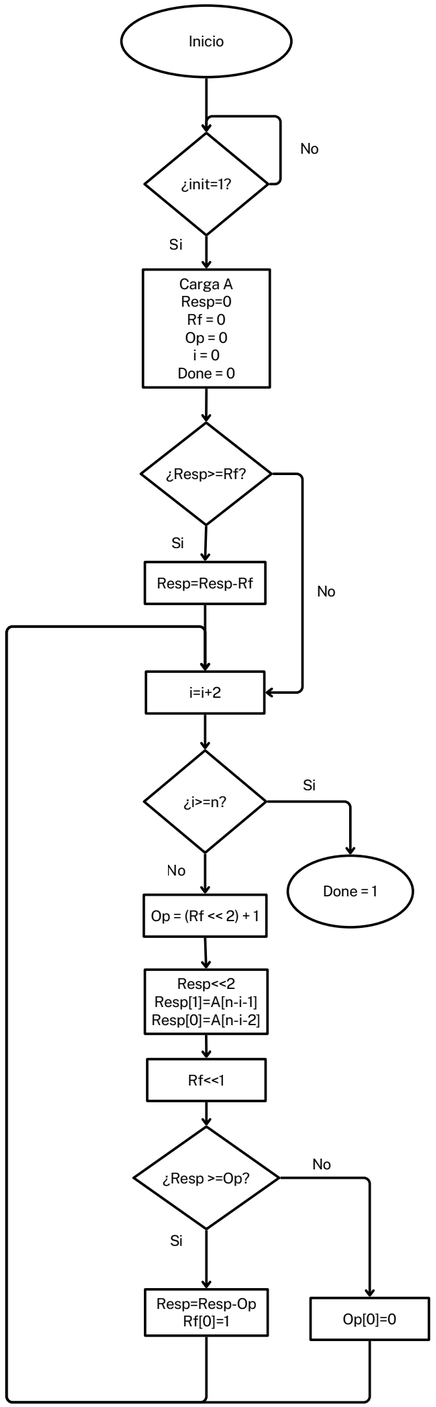
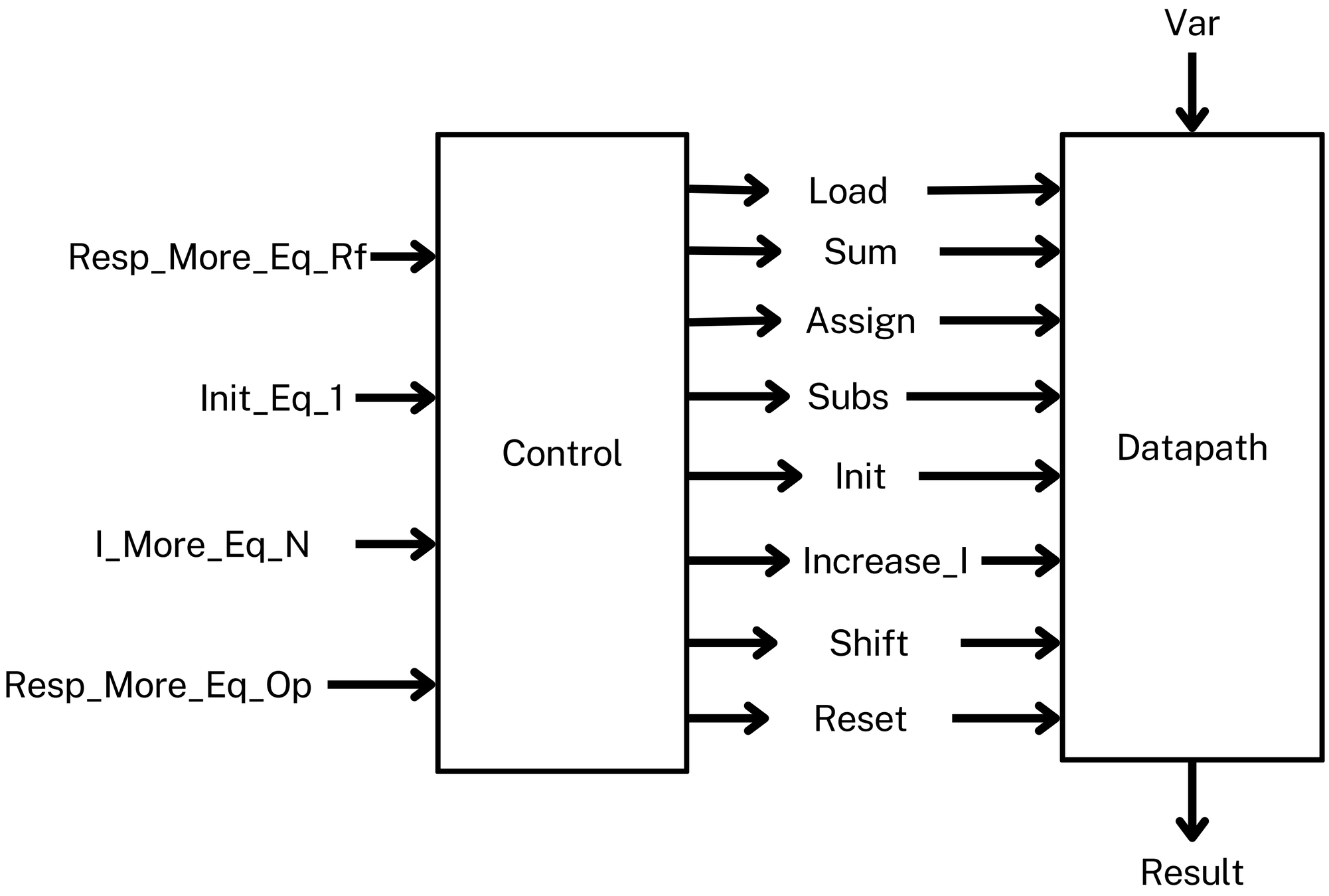
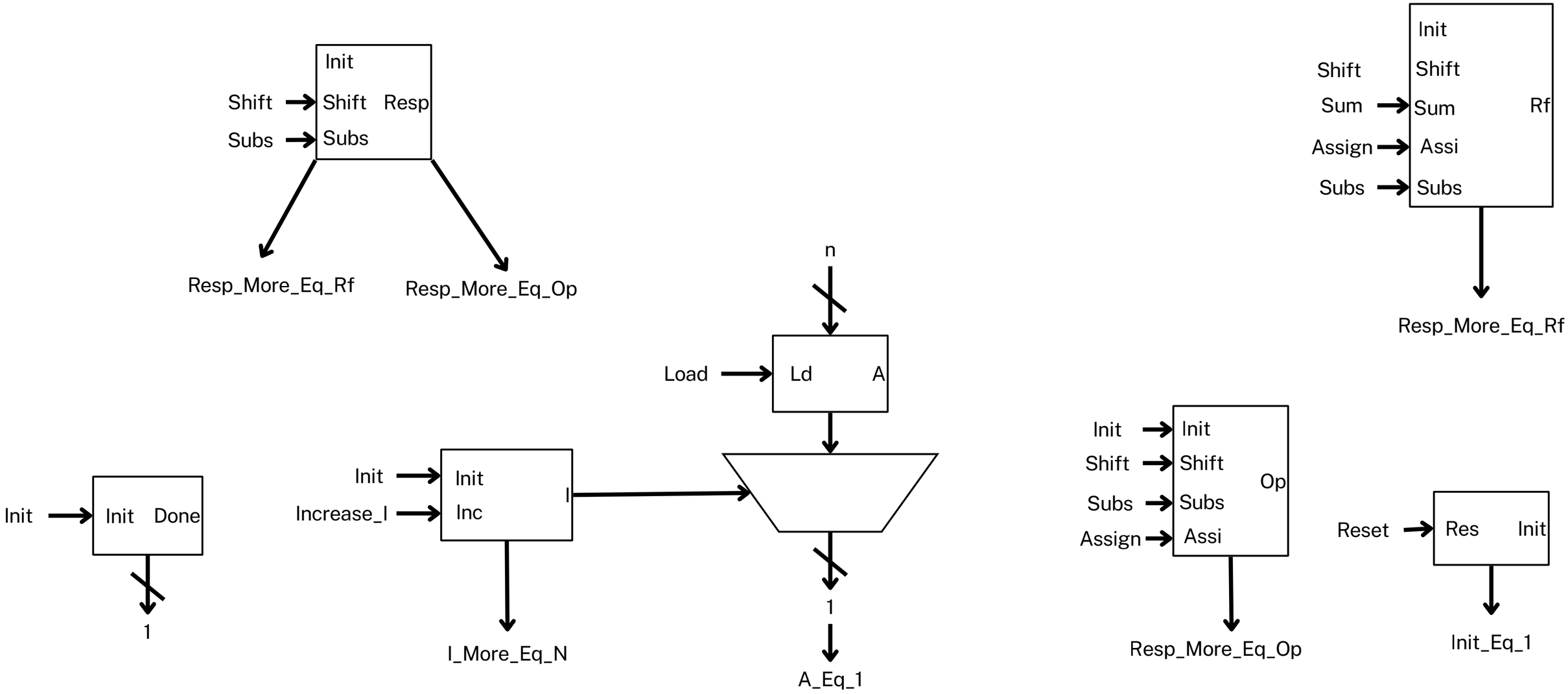
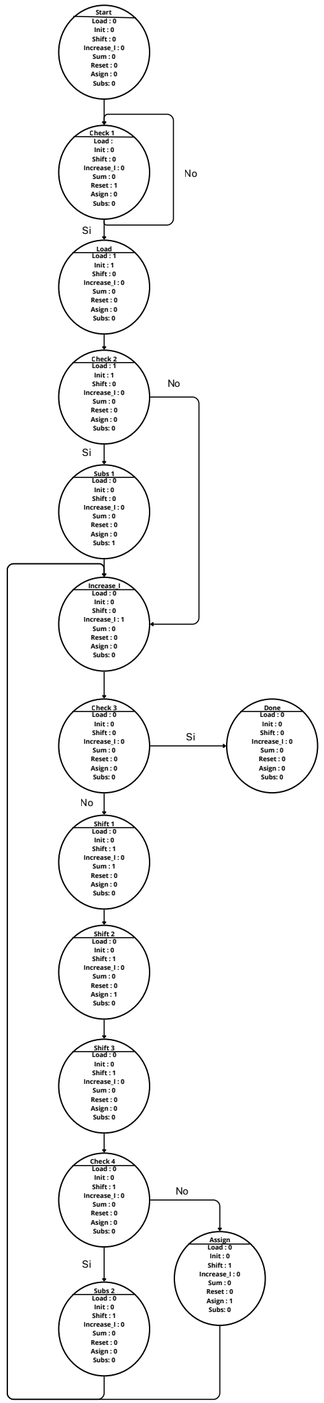

A
n
Init
1
Caja negra
1
Rf
1
Done


Inicio
¿init=1?
No
Si
Carga A
Resp=0
Rf = 0
Op = 0
i = 0
Done = 0
¿Resp>=Rf?
Si
Resp=Resp-Rf
No
i=i+2
¿i>=n?
Si
Done = 1
No
Op = (Rf << 2) + 1
Resp<<2
Resp[1]=A[n-i-1]
Resp[0]=A[n-i-2]
Rf<<1
¿Resp >=Op?
No
Si
Resp=Resp-Op
Rf[0]=1
Op[0]=0


Var
Resp_More_Eq_Rf
Init_Eq_1
I_More_Eq_N
Resp_More_Eq_Op
Control
Load
Sum
Assign
Subs
Init
Increase_I
Shift
Reset
Datapath
Result


Init
Shift
Shift
Resp
Subs
Subs
Resp_More_Eq_Rf
Resp_More_Eq_Op
Init
Shift
Shift
Sum
Sum
Rf
Assign
Assi
Subs
Subs
Resp_More_Eq_Rf
n
Load
Ld
A
Init
Init
Done
1
Init
Increase_I
Init
Inc
I
I_More_Eq_N
1
A_Eq_1
Init
Init
Shift
Shift
Op
Subs
Subs
Assign
Assi
Resp_More_Eq_Op
Reset
Res
Init
Init_Eq_1


Start
Load : 0
Init : 0
Shift : 0
Increase_I : 0
Sum : 0
Reset : 0
Asign : 0
Subs: 0
Check 1
Load :
Init : 0
Shift : 0
Increase_I : 0
Sum : 0
Reset : 1
Asign : 0
Subs: 0
No
Si
Load
Load : 1
Init : 1
Shift : 0
Increase_I : 0
Sum : 0
Reset : 0
Asign : 0
Subs: 0
Check 2
Load : 1
Init : 1
Shift : 0
Increase_I : 0
Sum : 0
Reset : 0
Asign : 0
Subs: 0
No
Si
Subs 1
Load : 0
Init : 0
Shift : 0
Increase_I : 0
Sum : 0
Reset : 0
Asign : 0
Subs: 1
Increase_I
Load : 0
Init : 0
Shift : 0
Increase_I : 1
Sum : 0
Reset : 0
Asign : 0
Subs: 0
Check 3
Load : 0
Init : 0
Shift : 0
Increase_I : 0
Sum : 0
Reset : 0
Asign : 0
Subs: 0
Si
Done
Load : 0
Init : 0
Shift : 0
Increase_I : 0
Sum : 0
Reset : 0
Asign : 0
Subs: 0
No
Shift 1
Load : 0
Init : 0
Shift : 1
Increase_I : 0
Sum : 1
Reset : 0
Asign : 0
Subs: 0
Shift 2
Load : 0
Init : 0
Shift : 1
Increase_I : 0
Sum : 0
Reset : 0
Asign : 1
Subs: 0
Shift 3
Load : 0
Init : 0
Shift : 1
Increase_I : 0
Sum : 0
Reset : 0
Asign : 0
Subs: 0
Check 4
Load : 0
Init : 0
Shift : 1
Increase_I : 0
Sum : 0
Reset : 0
Asign : 0
Subs: 0
No
Si
Assign
Load : 0
Init : 0
Shift : 1
Increase_I : 0
Sum : 0
Reset : 0
Asign : 1
Subs: 0
Subs 2
Load : 0
Init : 0
Shift : 1
Increase_I : 0
Sum : 0
Reset : 0
Asign : 0
Subs: 0